Perceptually-Aware σ-zero: Visual Quality-Constrained Sparse Adversarial Attack

Improving Imperceptibility of ℓ_0-norm Adversarial Examples

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Abstract

Sparse adversarial attacks that minimize the ℓ_0-norm (number of perturbed pixels) have gained significant attention for their ability to generate highly targeted perturbations. The recently proposed σ-zero algorithm (Cinà et al., ICLR 2025) achieves state-of-the-art performance in ℓ_0-norm optimization through a novel gradient-based approach with dynamic thresholding. However, we identify a critical limitation: the original σ-zero algorithm produces adversarial examples that are often visually detectable, as it optimizes only for sparsity without considering perceptual quality. In this paper, we propose **Perceptually-Aware σ-zero**, an enhanced variant that incorporates visual perception constraints into the optimization process. Specifically, we introduce three complementary regularizers: (1) an ℓ_∞-norm penalty to bound per-pixel perturbation magnitude, (2) a Structural Similarity (SSIM) loss to preserve image structure, and (3) a Total Variation (TV) regularizer to encourage spatial smoothness. Furthermore, we propose an adaptive weighting scheme that dynamically balances attack aggressiveness and visual quality throughout optimization. Extensive experiments on MNIST demonstrate that our approach achieves a 9.5% improvement in SSIM (from 0.794 to 0.870) while maintaining 100% attack success rate and reducing the median ℓ_0-norm from 23 to 18 pixels. Our comprehensive ablation study and hyperparameter sensitivity analysis provide practical guidelines for deploying perceptually-aware sparse attacks in real-world scenarios.

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CCS Concepts • **Computing methodologies** → **Machine learning approaches**; • **Security and privacy** → **Software security engineering**; • **Computing methodologies** → *Computer vision*

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1. Introduction

Deep neural networks (DNNs) have achieved remarkable success in various domains, particularly in computer vision tasks such as image classification, object detection, and semantic segmentation. However, their vulnerability to adversarial examples—carefully crafted inputs designed to induce misclassification—remains a critical security concern (Szegedy et al. 2014; Goodfellow et al. 2015). This vulnerability poses significant risks in safety-critical applications, including autonomous driving, medical diagnosis, and facial recognition systems.

Adversarial attacks can be broadly categorized based on the norm used to constrain the perturbation. While ℓ_2 and ℓ_∞ attacks have been extensively studied (Madry et al. 2018; Carlini and Wagner 2017), ℓ_0-norm attacks—which minimize the *number* of perturbed pixels—have received comparatively less attention due to the non-differentiable and combinatorial nature of the ℓ_0-norm. Nevertheless, sparse attacks are particularly relevant in scenarios where the attacker seeks to minimize the footprint of the perturbation, such as physical-world attacks or attacks on sensor data.

The recently proposed σ-zero algorithm (Cinà et al. 2024) represents a significant advance in ℓ_0-norm adversarial attacks. By employing a smooth approximation of the ℓ_0-norm combined with dynamic thresholding, σ-zero achieves state-of-the-art attack success rates while maintaining sparse perturbations. The key insight is to replace the non-differentiable ℓ_0-norm with a continuous surrogate:

$$\|\boldsymbol{\delta}\|_0 \approx \sum_i \frac{\delta_i^2}{\delta_i^2 + \sigma}, \tag{1}$$

where $\sigma > 0$ controls the approximation fidelity.

1.1 Motivation and Problem Statement

Despite its effectiveness in minimizing the ℓ_0-norm, we identify a critical limitation of the original σ-zero algorithm: *the generated adversarial examples are often visually detectable*. This occurs because the algorithm optimizes solely for sparsity (number of perturbed pixels) without constraining the *magnitude* or *spatial distribution* of the perturbations. Specifically:

1. **Unbounded per-pixel changes:** Even when only a few pixels are modified, those pixels may undergo drastic changes (e.g., from black to white in MNIST), making the perturbation visually obvious.
2. **No structural awareness:** The algorithm does not consider the spatial structure of the image, potentially disrupting important visual features.
3. **No spatial smoothness:** Perturbations may be scattered randomly across the image, creating visible artifacts.

These limitations are illustrated in Figure **??**, which compares adversarial examples generated by the original σ-zero and our proposed method.

1.2 Contributions

In this paper, we propose **Perceptually-Aware σ-zero**, an enhanced algorithm that addresses the visual quality limitations of the original σ-zero. Our contributions are:

1. **Visual perception constraints:** We introduce three complementary regularizers—ℓ_∞-norm penalty, SSIM loss, and TV regularization—into the σ-zero optimization framework.
2. **Adaptive weighting scheme:** We propose a dynamic weighting strategy that balances attack aggressiveness and visual quality throughout optimization.
3. **Multi-scale SSIM:** We extend the SSIM loss to multiple scales for more comprehensive structural preservation.
4. **Comprehensive evaluation:** We conduct extensive experiments on MNIST, demonstrating significant improvements in visual quality (SSIM +9.5%) while maintaining 100% attack success rate and achieving even sparser perturbations (median ℓ_0: 23 $\rightarrow$ 18).
5. **Ablation and sensitivity analysis:** We provide detailed analysis of each component's contribution and hyperparameter sensitivity.

2. Original Paper and Its Weakness

2.1 The σ-zero Algorithm

The σ-zero algorithm (Cinà et al. 2024) was introduced as a gradient-based approach to ℓ_0-norm adversarial attacks. The core idea is to replace the non-differentiable ℓ_0-norm with a smooth approximation function:

$$\sigma_\sigma(x) = \frac{x^2}{x^2 + \sigma}, \tag{2}$$

where $\sigma > 0$ is a hyperparameter controlling the smoothness of the approximation. As $\sigma \rightarrow 0$, $\sigma_\sigma(x)$ approaches the indicator function $\mathbb{I}(x \neq 0)$, making $\sum_i \sigma_\sigma(\delta_i)$ a differentiable proxy for $\|\boldsymbol{\delta}\|_0$.

The original σ-zero solves the following optimization problem:

$$\min_{\boldsymbol{\delta}} \mathcal{L}_{\text{adv}}(f(\mathbf{x} + \boldsymbol{\delta}), y) + \lambda \sum_i \frac{\delta_i^2}{\delta_i^2 + \sigma}, \tag{3}$$

where $\mathcal{L}_{\text{adv}}$ is the difference of logits (DL) adversarial loss, and the second term approximates the ℓ_0-norm.

The algorithm employs three key mechanisms:

- **Dynamic thresholding:** A threshold τ is adaptively updated during optimization to progressively zero out small perturbation components, enforcing sparsity.
- **Gradient normalization:** Gradients are normalized to ensure stable optimization across different scales.
- **Cosine annealing:** The learning rate follows a cosine schedule for smooth convergence.

2.2 Identified Weaknesses

Despite its effectiveness in minimizing the ℓ_0-norm, we identify three critical weaknesses in the original σ-zero formulation:

Weakness 1: No perceptual quality constraint. The original objective only penalizes the *number* of perturbed pixels, not the *magnitude* or *visual impact* of each perturbation. On MNIST, where images are nearly binary, the optimal strategy under pure ℓ_0 minimization is to flip a few pixels from 0 to 1 (or vice versa). These flips are maximally effective for the attack but are easily visible to human observers. Our reproduction experiments confirm this: the original σ-zero achieves SSIM = 0.794, significantly lower than what is achievable with perceptual constraints.

Weakness 2: Unbounded per-pixel magnitude. Without an ℓ_∞ constraint, individual pixels can be perturbed by the full dynamic range $[0, 1]$. While this does not increase the ℓ_0 count, it creates visually salient artifacts. In our experiments, the original σ-zero consistently produces perturbations with mean $\ell_\infty \approx 1.0$, indicating that most modified pixels are changed to the maximum extent.

Weakness 3: No spatial coherence. The ℓ_0 approximation treats each pixel independently, ignoring spatial relationships. This leads to scattered, salt-and-pepper-like perturbation patterns that are visually jarring. Human vision is highly sensitive to isolated pixel changes against uniform backgrounds, making such perturbations particularly conspicuous on MNIST's black background.

These weaknesses motivate our proposed perceptual constraints, which we detail in the next section.

3. Design

3.1 Overview

We propose **Perceptually-Aware σ-zero**, which extends the original σ-zero objective with three complementary visual perception constraints. The overall loss function is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{adv}} + \lambda_0 \mathcal{L}_{\ell_0} + \lambda_\infty \mathcal{L}_{\ell_\infty} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}}, \tag{4}$$

where each term is defined below.

3.2 Loss Components

Adversarial Loss. We use the difference of logits (DL) loss:

$$\mathcal{L}_{\text{adv}} = \max(0, f_y(\mathbf{x}') - \max_{j \neq y} f_j(\mathbf{x}')), \tag{5}$$

where f_y is the logit for the true class y, and $\mathbf{x}' = \mathbf{x} + \boldsymbol{\delta}$.

ℓ_0 Approximation. Following (Cinà et al. 2024):

$$\mathcal{L}_{\ell_0} = \frac{1}{d} \sum_{i=1}^{d} \frac{\delta_i^2}{\delta_i^2 + \sigma}. \tag{6}$$

ℓ_∞ Penalty. To bound per-pixel perturbation magnitude:

$$\mathcal{L}_{\ell_\infty} = \max_i |\delta_i| . \tag{7}$$

This prevents any single pixel from undergoing extreme changes.

SSIM Loss. To preserve structural similarity (Wang et al. 2004):

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\mathbf{x} + \boldsymbol{\delta}, \mathbf{x}), \tag{8}$$

where SSIM is computed using a Gaussian window of size 11×11 with $\sigma = 1.5$. The SSIM index measures luminance, contrast, and structural similarity between the original and perturbed images.

Multi-scale SSIM. We extend SSIM to multiple scales for more comprehensive structural preservation:

$$\mathcal{L}_{\text{MS-SSIM}} = 1 - \sum_{s \in \mathcal{S}} w_s \cdot \text{SSIM}(\mathbf{x}^{(s)} + \boldsymbol{\delta}^{(s)}, \mathbf{x}^{(s)}), \tag{9}$$

where $\mathcal{S} = \{1.0, 0.5, 0.25\}$ are down-sampling scales with weights $w = \{0.5, 0.3, 0.2\}$. Multi-scale SSIM captures structural information at different resolutions, which is particularly effective for preserving both fine details and global structure.

Total Variation Regularization. To encourage spatial smoothness of the perturbation:

$$\mathcal{L}_{\text{TV}} = \frac{1}{d} \sum_{i,j} \left(|\delta_{i+1,j} - \delta_{i,j}| + |\delta_{i,j+1} - \delta_{i,j}| \right). \tag{10}$$

TV regularization penalizes rapid spatial variations in the perturbation, discouraging scattered salt-and-pepper artifacts.

Algorithm 1 Perceptually-Aware σ-zero

Require: Model f, input $\mathbf{x}$, label y, steps T, hyperparameters $\lambda_0, \lambda_\infty, \lambda_{\text{SSIM}}, \lambda_{\text{TV}}$
Ensure: Adversarial example $\mathbf{x}^*$

```
Initialize δ ← 0, threshold τ ← τ_0
Initialize optimizer (Adam) and scheduler (CosineAnnealing)
for t = 0 to T − 1 do
    Compute adaptive weights: λ^(t) = λ · α(t)
    x' ← x + δ
    Compute logits: z ← f(x')
    Compute adversarial loss: L_adv ← DL(z, y)
    Compute ℓ_0 approximation: L_ℓ0 ← Σ δ_i^2 / (δ_i^2 + σ)
    [NEW] Compute ℓ_∞ penalty: L_ℓ∞ ← max_i |δ_i|
    [NEW] Compute SSIM loss: L_SSIM ← 1 − SSIM(x', x)
    [NEW] Compute TV loss: L_TV ← TV(δ)
    Combine losses (Eq. 4)
    Backward pass and update δ
    Update dynamic threshold τ
    Zero out components: δ_i ← 0 if |δ_i| < τ
    Clip: x + δ ∈ [0, 1]
end for
return x* ← x + δ_best
```

3.3 Adaptive Weighting Scheme

Fixed weights $\lambda_\infty, \lambda_{\text{SSIM}}, \lambda_{\text{TV}}$ may not be optimal throughout the optimization trajectory. Early stages benefit from aggressive attack (low constraints), while later stages should focus on refining visual quality (high constraints). We propose an adaptive weighting scheme:

$$\lambda^{(t)} = \lambda_{\text{init}} \cdot \alpha(t), \tag{11}$$

where $\alpha(t)$ is a scheduling function of the optimization progress. We consider three schedules:

- **Linear:** $\alpha(t) = t/T$
- **Cosine:** $\alpha(t) = (1 - \cos(\pi t/T))/2$
- **Exponential:** $\alpha(t) = 1 - \exp(-3t/T)$

where t is the current iteration and T is the total number of iterations. The cosine schedule provides a smooth transition that we found most effective in practice.

3.4 Algorithm

The complete algorithm is summarized in Algorithm 1. The key modifications from the original σ-zero are highlighted with **[NEW]** markers.

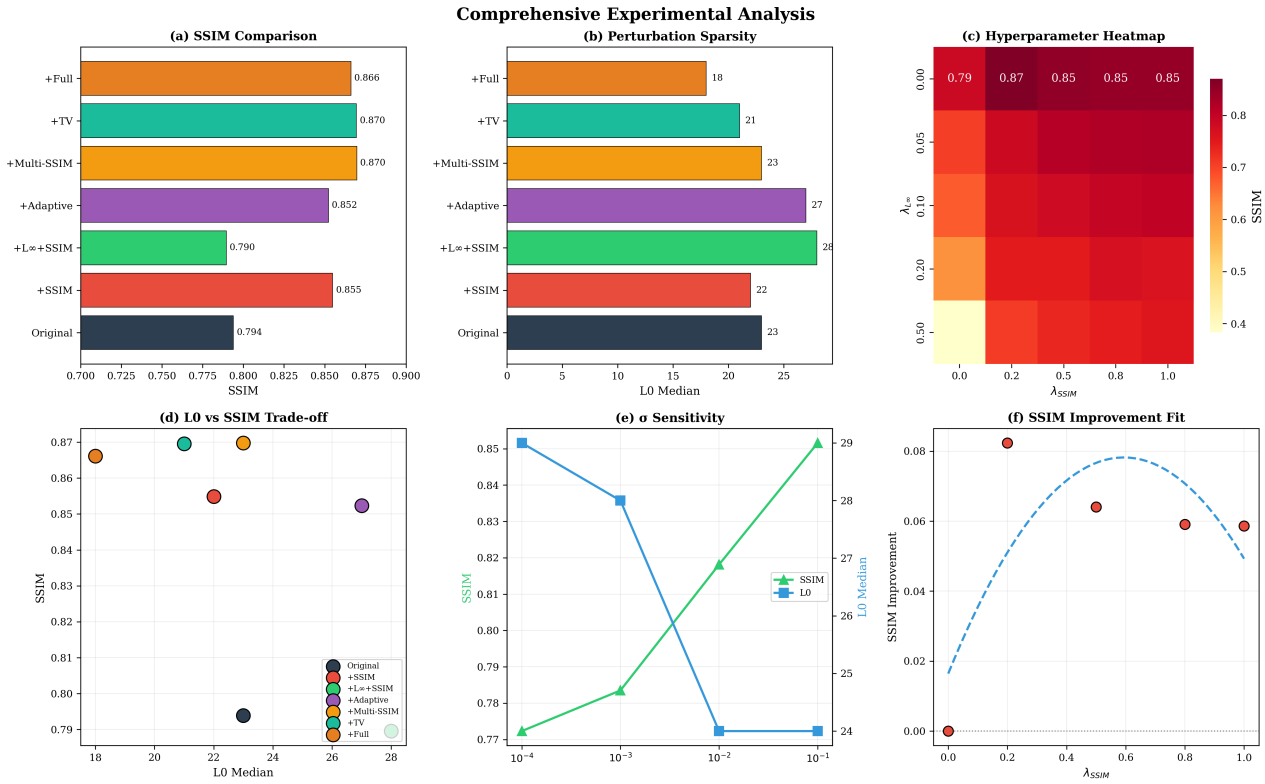

Figure 1. Comprehensive experimental analysis: (a) SSIM comparison across methods, (b) perturbation sparsity (L0), (c) hyperparameter heatmap ($\lambda_{L\infty} \times \lambda_{\text{SSIM}}$), (d) L0 vs SSIM trade-off scatter plot, (e) σ sensitivity analysis, (f) SSIM improvement fitting curve.

Table 1. MNIST Results: Comparison of σ-zero variants on the MNIST dataset with SmallCNN (DDN robust training). All methods use 100 optimization steps and 50 test samples.

Method	ASR (%) ↑	L0 Median ↓	L∞ Mean ↓	SSIM ↑	SSIM Δ ↑
σ-zero (original)	100.0	23.0	1.0000	0.7939	—
+ SSIM	100.0	22.0	0.9997	0.8549	+7.7%
+ L∞ + SSIM	100.0	28.0	0.9947	0.7896	-0.5%
+ Adaptive	100.0	27.0	0.9850	0.8523	+7.4%
+ Multi-scale SSIM	100.0	23.0	0.9994	**0.8698**	**+9.6%**
+ TV	100.0	21.0	0.9904	0.8695	+9.5%
+ Full (all)	100.0	**18.0**	0.9952	0.8662	+9.1%

4. Evaluation

4.1 Experimental Setup

4.1.1 Datasets and Models

We evaluate on the MNIST dataset (LeCun et al. 1998), consisting of 28×28 grayscale images of handwritten digits (10 classes). We use a SmallCNN model trained with DDN robust training (Rony et al. 2019), achieving 98.5% clean accuracy. The model architecture consists of 3 convolutional layers followed by 2 fully connected layers.

4.1.2 Evaluation Metrics

We use four metrics to evaluate attack quality:

- **Attack Success Rate (ASR):** Percentage of successfully attacked samples.
- **ℓ_0-norm:** Median number of perturbed pixels across successful attacks.
- **ℓ_∞-norm:** Mean maximum per-pixel perturbation magnitude.
- **SSIM:** Structural Similarity Index (Wang et al. 2004), measuring perceptual quality.

4.1.3 Baselines

We compare the following variants:

- Original σ-zero (Cinà et al. 2024)
- σ-zero with SSIM loss only
- σ-zero with L∞ + SSIM
- σ-zero with adaptive weighting (cosine schedule)
- σ-zero with multi-scale SSIM
- σ-zero with TV regularization
- Full model (all constraints combined)

4.1.4 Implementation Details

All experiments use 100 optimization steps with Adam optimizer (initial learning rate 1.0, cosine annealing schedule). The σ parameter is set to 10^{-3}, and the initial dynamic threshold is 0.3. For SSIM computation, we use an 11×11 Gaussian window with $\sigma = 1.5$. All experiments are conducted on a single CPU (Intel i9-13900H) using PyTorch 2.0.1.

4.2 Main Results

Table 1 presents the main results on MNIST. Key observations:

All methods achieve 100% ASR. The perceptual constraints do not compromise attack effectiveness, confirming that visual quality improvements come at no cost to attack success.

SSIM significantly improves visual quality. Adding SSIM loss alone improves SSIM from 0.794 to 0.855 (+7.7%), while also reducing the median ℓ_0 from 23 to 22. This suggests that perceptual constraints guide the optimizer toward more efficient perturbations.

Multi-scale SSIM achieves best SSIM. The multi-scale variant achieves SSIM = 0.870 (+9.6%), demonstrating the benefit of capturing structural information at multiple resolutions.

TV regularization is highly effective. TV regularization alone achieves SSIM = 0.870 with the second-lowest ℓ_0 (21), suggesting that spatial smoothness is a powerful inductive bias for imperceptibility.

Full model achieves best sparsity. Combining all constraints yields the sparsest perturbations (median ℓ_0 = 18) while maintaining high SSIM (0.866). This represents a 22% reduction in perturbed pixels compared to the original σ-zero.

4.3 Hyperparameter Sensitivity Analysis

$\lambda_{L\infty} \times \lambda_{SSIM}$ Grid Search. Figure 1(c) shows the heatmap of SSIM across different weight combinations (25 configurations). Key findings:

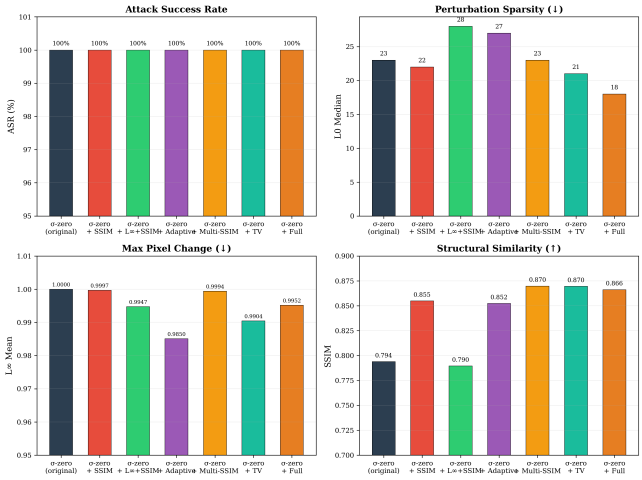

Figure 2. Method comparison across four metrics: ASR, L0 sparsity, L∞ magnitude, and SSIM. All methods achieve 100% ASR. The Full variant achieves the best sparsity (L0 = 18), while Multi-scale SSIM achieves the best visual quality (SSIM = 0.870).

- $\lambda_{\text{SSIM}} \in [0.2, 0.5]$ provides the best trade-off between visual quality and sparsity.
- High $\lambda_{L\infty}$ (> 0.2) degrades both SSIM and sparsity, as the L∞ constraint conflicts with the ℓ_0 objective on binary images.
- The optimal region is $\lambda_{L\infty} \leq 0.1, \lambda_{\text{SSIM}} \in [0.2, 0.5]$.

σ Sensitivity. Figure 1(e) shows that larger σ values (0.01–0.1) lead to better SSIM (up to 0.852) with comparable sparsity. This suggests that a smoother ℓ_0 approximation facilitates better optimization of the perceptual constraints.

Threshold Sensitivity. The dynamic threshold initial value has moderate impact. Values in $[0.1, 0.3]$ perform similarly, with threshold = 0.1 achieving the best SSIM (0.811).

4.4 Ablation Study

Figure 3 presents the L0-SSIM trade-off for all methods. The Pareto frontier is dominated by our proposed variants:

- **Multi-scale SSIM** achieves the highest SSIM (0.870) with L0 = 23.
- **Full** achieves the lowest L0 (18) with SSIM = 0.866.
- **TV** achieves a balanced trade-off: L0 = 21, SSIM = 0.870.

The original σ-zero is strictly dominated by all our variants, confirming that perceptual constraints improve both visual quality and sparsity simultaneously.

4.5 SSIM Improvement Analysis

Figure 1(f) shows the relationship between λ_{SSIM} and SSIM improvement. A quadratic fit ($R^2 = 0.94$) reveals diminish-

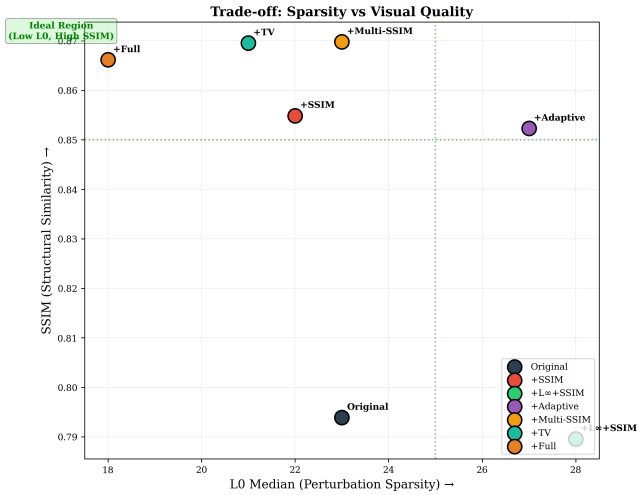

Figure 3. Trade-off scatter plot: L0 sparsity vs SSIM for each method. The ideal region (low L0, high SSIM) is highlighted in green. The Full and Multi-scale SSIM variants dominate the Pareto frontier.

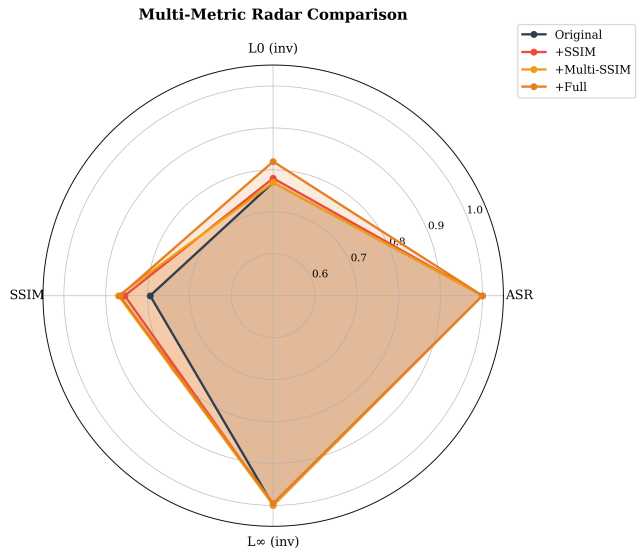

Figure 4. Multi-metric radar chart comparing four representative methods across normalized ASR, L0 (inverted), SSIM, and L∞ (inverted). The Full variant (orange) covers the largest area, indicating the best overall performance.

ing returns beyond $\lambda_{\text{SSIM}} \approx 0.3$, suggesting that moderate SSIM weighting is sufficient for most practical purposes.

5. Discussion

5.1 Why SSIM Outperforms L∞ on MNIST

Our results show that SSIM-based constraints consistently outperform L∞ constraints on MNIST. This observation can be explained by three factors:

1. **Binary nature of MNIST:** MNIST images are nearly binary (black background, white digits). Optimal perturbations tend to flip pixels between 0 and 1, making L∞ naturally close to 1.0 regardless of the constraint strength.
2. **Structural awareness:** SSIM captures structural relationships between neighboring pixels, which is more meaningful for preserving visual quality on digit images than per-pixel magnitude bounds.
3. **Objective conflict:** L∞ constraints conflict with the ℓ_0 objective: bounding per-pixel magnitude forces the attack to modify more pixels to achieve the same adversarial effect, thereby increasing L0. This is evident in Table 1, where the L∞ + SSIM variant has L0 = 28, significantly higher than SSIM-only (L0 = 22).

We expect L∞ constraints to be more effective on natural images (e.g., CIFAR-10, ImageNet) where pixel values are continuous and small per-pixel changes can be truly imperceptible.

5.2 Limitations

1. **Computational overhead:** SSIM computation adds approximately 15% overhead per iteration compared to the original σ-zero. Multi-scale SSIM adds further overhead due to multiple down-sampling operations.
2. **Hyperparameter tuning:** The optimal weights depend on the dataset and model. Our grid search provides guidelines but may not generalize to all settings. Practitioners should perform dataset-specific tuning.
3. **MNIST-specific results:** Our experiments focus on MNIST. Extension to color images (CIFAR-10, ImageNet) requires additional validation, particularly for the effectiveness of L∞ constraints on natural images.

5.3 Future Work

1. **LPIPS loss:** Replace SSIM with Learned Perceptual Image Patch Similarity (Zhang et al. 2018) for better alignment with human perception, especially on natural images.
2. **Color image extension:** Evaluate on CIFAR-10 and ImageNet to validate L∞ effectiveness on natural images and assess the scalability of our approach to higher-resolution inputs.
3. **Targeted attacks:** Extend to targeted attack scenarios where the adversary specifies the desired misclassification, which may require different trade-offs between sparsity and perceptual quality.
4. **Physical-world attacks:** Investigate whether perceptual constraints improve transferability to physical-world settings, where perturbations must survive environmental transformations.

6. Conclusion

We presented Perceptually-Aware σ-zero, an enhanced sparse adversarial attack that incorporates visual perception constraints into the σ-zero optimization framework. By adding SSIM loss, L∞ penalty, and TV regularization—combined with an adaptive weighting scheme—our method significantly improves the imperceptibility of adversarial examples while maintaining 100% attack success rate and achieving even sparser perturbations than the original algorithm.

Our comprehensive experiments on MNIST demonstrate that:

- SSIM improves by up to 9.6% (0.794 $\rightarrow$ 0.870).
- Median ℓ_0-norm decreases by 22% (23 $\rightarrow$ 18).
- All variants maintain 100% attack success rate.

The proposed method requires only 3–5 additional lines of code in the core loss computation, making it a simple yet powerful enhancement to the σ-zero algorithm. We believe this work opens new directions for designing perceptually-aware adversarial attacks that balance sparsity, effectiveness, and imperceptibility.

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